

JKBOSE 2026

CLASS 11TH

Plant Growth and Development

LECTURE-05

BOTANY

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Topics To Covered

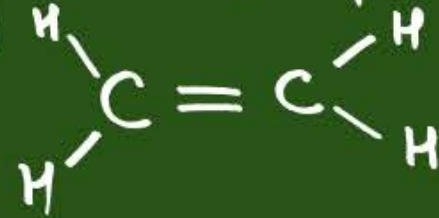
- 1** Ethylene
- 2** Absciscic Acid (ABA)

Ethylene

≡ Gaseous per

≡ It is most widely used per in agriculture.

≡ C_2H_4 (ethylene)



≡ Precursor: Methionine.

Discovery

H.H Cousins (1910)

↓
observed ripening of bananas by a volatile gas
secreted by ripened oranges.

→ later on this volatile gas was identified as Ethylene.

Role of Ethylene

- ① It promotes Senescence of fruits, leaves, flowers.
- ② It promotes ripening of fruits. During ripening, rate of respiration increases.
- ③ It shows triple response
 - a. Horizontal growth of axis.
 - b. Swelling / Broadening of axis.
 - c. Apical hook formation

in Dicot Seedling



- ④. Promotes root growth and root hair formation and thus increases surface area of roots for absorption of H_2O and minerals.
- ⑤. Promotes elongation of internodes and petioles in deep water rice-seedling plants.
- ⑥. Promotes flowering in pineapple, Mango.
- ⑦. It breaks bud dormancy in potato and thus helps in sprouting of potato.
- ⑧. It breaks seed dormancy in peanut.
- ⑨. It causes synchronization of fruit-set in pineapple.

ETHEPHON

- Aqueous form of Ethylene.
- Easily absorbed by plants through roots.
- It gets readily transported to all the plant parts.

⊗ Ethephon increases number of female flowers (i.e., females) in cucumber and thus increases their yield.

⊗ promotes ripening of fruits (Tomato, Apple).

⊗ promotes abscission of leaves, fruits, flowers and thus causes thinning ex. Cotton
Walnut
Cherry.

Abscissic Acid (ABA)



- AKA Stress hormone.

Discovery

→ In 1960, 03 Chemicals were discovered independently

→ Abscission-II
→ Dormin
→ Inhibitor-B

} later on it was discovered that all these three chemical were chemically identical and then named as Abscissic Acid (ABA)

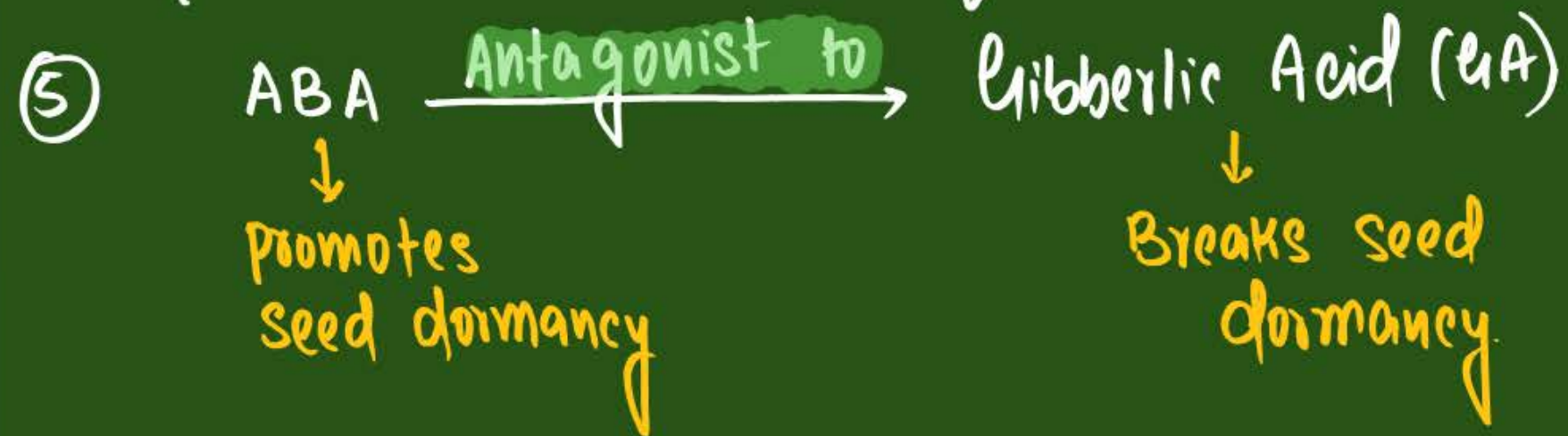
Precursor: Carotenoids.

Site:- Seed, Stomata, leaf.

Role of Abscissic Acid (ABA)



- ① It is general plant growth inhibitor.
- ② Promotes closure of Stomata (during H_2O Stress).
- ③ Promotes seed dormancy.
- ④ Promotes Abscission of leaves.



QUESTIONS

The gaseous plant growth regulator is used in plants to: [NEET-2022 (P-I)]

- 1 help overcome apical dominance
- 2 kill dicotyledonous weeds in the fields
- 3 speed up the malting process
- ✓ 4 promote root growth and root hair formation to increase the absorption surface

QUESTIONS

Production of Cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants:

[NEET-2022 (P-I)]

- ☒ 1 Ethylene
- ☐ 2 Cytokinin
- ☐ 3 ABA
- ☐ 4 Gibberellin

HOMEWORK

Revision



Syllabus completed



Thank
You